Beyond the Ticket: Identifying High-Impact Generative AI Opportunities in Customer Support

Executive Project Report

Executive Summary

Generative Artificial Intelligence (GenAI) is rapidly transforming customer support by enabling organizations to automate repetitive tasks, improve response times, and reduce operating costs. However, many companies still deploy AI uniformly across all customer support activities without considering that different workflows vary significantly in complexity, business risk, and automation potential.

This project develops a data-driven framework for determining where GenAI should be deployed within Flipkart's customer support organization. Rather than asking whether AI should replace support agents, the analysis identifies which workflows should be fully automated, which should be AI-assisted, and which should remain under human control.

Using approximately **85,000 customer support interactions**, the project combines descriptive analytics, workflow intelligence, machine learning, and Explainable AI (SHAP) to produce a practical roadmap for AI implementation. The final outcome is a workflow-centric strategy that prioritizes operational efficiency while maintaining customer experience and minimizing business risk.

Business Context

The Challenge

Customer support organizations are under increasing pressure to:

- Reduce operating costs
- Improve response times
- Increase customer satisfaction
- Scale operations without proportionally increasing headcount

Generative AI appears to provide a solution to these challenges. However, simply deploying AI-powered chatbots across every support workflow rarely produces the expected business value.

The Problem with Current AI Adoption

Many organizations treat customer support as a single business process.

In reality, customer support consists of dozens of different operational workflows, each requiring different levels of:

- Human judgment
- Operational coordination
- Business knowledge
- Decision-making

For example, checking an order status requires a completely different process than investigating fraud or resolving a seller dispute. Applying identical automation strategies across these fundamentally different workflows often results in poor customer experiences and inefficient AI investments.

Project Objective

The primary objective of this project was to determine **where Generative AI can create the greatest operational value within Flipkart's customer support organization.**

To achieve this objective, the project aimed to:

- Understand how customer support demand is distributed across workflows.
- Identify the operational processes associated with poor customer satisfaction.
- Develop a structured AI implementation framework based on business characteristics.
- Validate the proposed strategy using supervised machine learning.
- Translate analytical findings into practical business recommendations.

Ultimately, every stage of the project was designed to answer one strategic question:

Where should Flipkart invest in Generative AI to maximize operational efficiency while improving customer experience?

Dataset Overview

The analysis was performed using approximately **85,000 customer support tickets** containing structured operational information, including:

- Ticket Category
- Operational Sub-category
- Communication Channel
- Customer Satisfaction (CSAT)
- Customer Remarks
- Agent-related attributes
- Operational identifiers

Unlike many customer satisfaction datasets, this dataset combines operational process information with customer outcomes, making it possible to evaluate not only how customers feel but also which business processes are responsible for those experiences.

Analytical Approach

Rather than relying on a single analytical technique, the project followed an end-to-end analytics workflow in which each stage informed the next.

The overall methodology consisted of:

1. Data Quality Assessment
2. Exploratory Hypothesis Testing
3. Operational Workflow Analysis
4. Customer Satisfaction Analysis
5. AI Workflow Prioritization
6. Machine Learning Validation
7. Explainable AI (SHAP)
8. Strategic Business Recommendations

Each stage contributed evidence supporting the final AI implementation strategy.

Initial Data Assessment

The project began by evaluating the quality and completeness of the available data. Missing values, variable types, duplicates, and overall dataset structure were examined to determine whether the data was suitable for downstream analysis.

During this assessment, customer remarks initially appeared to be a promising source of information for Natural Language Processing (NLP). However, rather than immediately applying NLP techniques, the project first evaluated whether the text actually contained sufficient information to justify that approach.

This evidence-based methodology became one of the defining characteristics of the project and laid the foundation for the analytical decisions that followed.

Key Findings

Assessing the Feasibility of Text Analytics

At the outset of the project, customer remarks appeared to be a promising candidate for Natural Language Processing (NLP). Rather than immediately building sentiment models, an exploratory analysis was conducted to evaluate whether the available text contained sufficient information for meaningful language-based analytics.

Several text characteristics—including word count, character count, sentence count, punctuation usage, and capitalization patterns—were examined to understand the richness of the customer remarks field.

The analysis revealed that the vast majority of customer comments were either empty or contained only a few words. This lack of linguistic variation meant that any sentiment analysis would primarily distinguish between tickets containing comments and those containing no comments, rather than accurately measuring customer frustration or complaint severity.

Key Insight

Rather than forcing NLP into the project because text data existed, the analysis pivoted toward structured operational variables. This ensured that all subsequent analyses were based on the strongest and most reliable information available.

Understanding Customer Support Operations

Ticket Volume Distribution

The next stage of the analysis focused on understanding how customer support demand was distributed across the organization.

The results showed that customer support demand was highly concentrated rather than evenly distributed. A relatively small number of operational categories generated the majority of customer interactions, with **Returns** and **Order Related**issues accounting for the largest share of support tickets.

This finding suggested that improvements within only a handful of operational workflows could influence a significant proportion of overall customer support activity.

Workflow-Level Analysis

While category-level summaries provided a useful overview, analyzing individual workflows produced much richer operational insights.

The analysis identified recurring workflows such as:

- Return Requests
- Delayed Deliveries
- Installation and Demonstration Requests
- Refund Enquiries
- Reverse Pickup Requests
- Seller Cancellation Issues
- Product Information Requests
- Order Status Enquiries

Examining support requests at this level proved particularly valuable because AI implementation decisions are ultimately made at the workflow level rather than at the category level.

Key Insight

Not all workflows require the same level of automation. Some follow highly standardized procedures, while others depend heavily on operational coordination or human judgment.

Customer Satisfaction Analysis

Understanding ticket volume alone does not indicate where AI should be deployed. Consequently, customer satisfaction was analyzed across categories and operational workflows to identify where customers experienced the greatest operational friction.

The analysis demonstrated that customer dissatisfaction was concentrated within specific operational processes rather than evenly distributed across customer support.

Workflows associated with logistics and fulfillment—including delayed deliveries, installation scheduling, seller cancellations, and refund processing—consistently exhibited lower customer satisfaction than routine informational requests.

Conversely, several high-volume informational workflows maintained relatively strong customer satisfaction despite processing large numbers of customer interactions.

Key Insight

High ticket volume does not necessarily correspond to poor customer experience. AI investment decisions should therefore consider both operational demand and customer satisfaction rather than relying on either metric independently.

AI Workflow Prioritization Framework

From Analysis to Strategy

The descriptive analyses provided an understanding of where customer demand and dissatisfaction were concentrated. However, they did not directly answer the central business question of where AI should be implemented.

To bridge this gap, a business-driven AI Workflow Prioritization Framework was developed.

Rather than ranking workflows solely according to ticket volume, each operational process was evaluated using several business-oriented criteria:

- Operational complexity
- Required human judgment
- Process standardization
- Business and operational risk

Based on these characteristics, each workflow was assigned to one of four implementation strategies.

AI Recommendation Categories

Fully Automate

Routine, highly standardized workflows with minimal business risk.

Examples include order status enquiries, invoice requests, and product information requests.

AI-Assisted

Operational workflows that benefit from AI support while continuing to require human oversight.

Examples include delayed deliveries, refund processing, installation scheduling, and reverse pickup requests.

Human-Led

Workflows requiring significant contextual reasoning or customer-specific decision-making.

AI serves as an assistant rather than the decision-maker.

Requires Investigation

High-risk workflows involving fraud, seller disputes, or complex operational investigations where human expertise remains essential.

Strategic Insight

One of the most significant findings of this project is that **the greatest opportunity for Generative AI lies in augmenting operational workflows rather than simply automating customer conversations**.

Instead of replacing customer support representatives, AI should function as a productivity tool that reduces repetitive work, accelerates information retrieval, improves operational visibility, and enables agents to resolve customer issues more efficiently.

This section establishes the business case. The **final chunk** will show that these recommendations were independently validated using machine learning and Explainable AI before concluding with the strategic recommendations and overall project conclusion. That creates a strong narrative: **business problem → analysis → strategy → validation → recommendations**.

Machine Learning Validation

Purpose

The AI Workflow Prioritization Framework developed during the earlier stages of the project was primarily based on descriptive analytics and business reasoning. To increase confidence in these recommendations, a supervised machine learning model was developed to determine whether the same operational characteristics identified during exploratory analysis also emerged as statistically significant predictors of poor customer satisfaction.

Rather than treating machine learning as the primary objective of the project, predictive analytics served as an independent validation mechanism for the proposed AI strategy.

Model Development

Customer satisfaction ratings were transformed into a binary classification problem, allowing support interactions to be classified as either satisfactory or at risk of generating poor customer experiences.

A LightGBM classifier was selected because of its strong performance on structured tabular datasets, computational efficiency, and compatibility with Explainable Artificial Intelligence techniques.

The model was developed using:

- Train-test split for unbiased evaluation
- Five-fold cross-validation during hyperparameter tuning
- Multiple classification metrics, including ROC-AUC, Precision, Recall, and F1-score

Interestingly, the baseline LightGBM model performed marginally better than the tuned model on the unseen test dataset. Rather than selecting the more complex model simply because additional tuning had been performed, the baseline model was retained for the remainder of the analysis.

Key Insight

Model selection should prioritize generalization performance rather than complexity. Hyperparameter tuning does not always improve predictive performance.

Explainable Artificial Intelligence (SHAP)

While predictive accuracy demonstrates that operational characteristics influence customer satisfaction, it does not explain *why* individual predictions are made.

To address this limitation, SHAP (SHapley Additive Explanations) was used to interpret the final LightGBM model.

The SHAP analysis identified the operational features contributing most strongly to customer dissatisfaction and provided both global and individual explanations for model predictions.

Several workflows identified earlier in the project—including delayed deliveries, installation and demonstration requests, seller cancellation issues, and return-related processes—also emerged as influential predictors within the machine learning model.

Key Insight

The predictive model independently validated the earlier descriptive findings, providing additional confidence that the proposed AI Workflow Prioritization Framework reflects genuine operational patterns rather than coincidental relationships within the dataset.

Strategic Recommendations

Based on the combined findings from descriptive analytics, workflow analysis, machine learning, and Explainable AI, the following implementation strategy is recommended.

Recommendation 1 – Automate Routine Requests

Fully automate highly standardized workflows that require minimal human judgment, including:

- Order status enquiries
- Invoice requests
- Product information requests
- Basic account-related requests

These workflows offer immediate efficiency gains while presenting relatively low implementation risk.

Recommendation 2 – Augment Operational Workflows

Deploy AI as a decision-support tool for workflows involving operational coordination rather than replacing support agents entirely.

Priority areas include:

- Delayed deliveries
- Refund processing
- Installation scheduling
- Reverse pickup requests

In these scenarios, AI should retrieve information, summarize customer history, recommend responses, and proactively communicate operational updates while allowing agents to retain decision-making authority.

Recommendation 3 – Preserve Human Expertise

Continue relying on human specialists for workflows involving significant business or financial risk.

These include:

- Fraud investigations
- Seller disputes
- Exceptional refund decisions
- Policy exceptions

AI should support these interactions by organizing information and reducing administrative effort rather than making autonomous decisions.

Recommendation 4 – Shift Toward Proactive Support

One of the most significant findings of this project is that many customer interactions are driven by uncertainty surrounding operational processes rather than failures within customer support itself.

Instead of focusing solely on answering customer questions, Flipkart should leverage AI to proactively communicate:

- Delivery updates
- Installation schedules
- Refund progress
- Reverse pickup confirmations
- Seller status notifications

Reducing unnecessary customer contacts represents a greater long-term opportunity than simply automating conversations.

Project Limitations

Several limitations should be considered when interpreting the findings.

- Customer remarks contained limited textual information, restricting the use of advanced NLP techniques.
- Certain operational metrics, including handling time and financial impact, were unavailable, requiring scenario-based business recommendations.
- The machine learning model was designed to validate business insights rather than function as a production-ready prediction system.
- The AI prioritization framework combines analytical evidence with business judgment and should therefore be refined alongside operational stakeholders before implementation.

Conclusion

This project demonstrates how business analytics can be used to guide strategic Artificial Intelligence investment decisions within customer support operations.

Rather than approaching AI adoption as a technology implementation exercise, the analysis evaluated customer support from an operational perspective, identifying where customer demand is concentrated, where dissatisfaction occurs, and which workflows are best suited for different levels of automation.

By integrating descriptive analytics, workflow intelligence, supervised machine learning, and Explainable Artificial Intelligence, the project developed a structured AI Workflow Prioritization Framework that balances operational efficiency, customer experience, and implementation risk.

The central conclusion is clear: **the greatest value of Generative AI lies not in replacing customer support agents with chatbots, but in redesigning customer support operations through workflow-specific automation, intelligent decision support, and proactive operational communication.**

This project provides a practical, evidence-based roadmap for organizations seeking to implement AI strategically rather than universally, demonstrating how analytics can transform operational data into actionable business decisions.